Hyperspectral Anomaly Detection Based on Two-Stage Deep Transformation With Latent Features Guidance

Maoyuan Feng, Yunxiu Yang, Yi Gan, Qin Shu, and Xiaoguang Shao

***Abstract*—Hyperspectral anomaly detection (HAD) has attracted widespread attention in the field of hyperspectral image processing. In recent years, model-driven and data-driven methods have been widely used in HAD. However, existing methods either do not fully utilize the various data characteristics of the background or directly use the various coupled data characteristics of the background in the image domain or a single deep transform domain, where the coupling characteristics in these domains are not obvious and difficult to separate, resulting in poor detection performance. To solve this problem, we innovatively proposed a two-stage deep transformation network with latent features guidance (TDTLFG) for HAD, aiming to fully utilize the potential data characteristics of the background tensor in different transform domains, and train the network under purposeful and directional guidance with that latent features guidance. Specifically, a multiscale feature extraction (MSFE) and a multiscale feature fusion (MSFF) two-stage deep transform blocks are designed to adapt to the anomaly scale-varying scenarios; the tensor matrix factorization is embedded into the network after the first MDFE deep transformation stage to guide the network to exploit the low rankness of the background in 3-D; the total variation (TV) constraint is designed on the output of the second MSFF transformation stage to exploit the piecewise-smoothness of the background in 3-D. After that, the fidelity constraints and TV constraints of the network output are iteratively solved by adopting the alternating direction method of multipliers (ADMM) algorithm. Finally, the detection performance effectiveness of the proposed model is proven on several real hyperspectral datasets.**

***Index Terms*—Deep learning, hyperspectral anomaly detection (HAD), tensor matrix factorization, total variation (TV).**

NOMENCLATURE

x	Scalar.
X	Matrix.
$\|x\|$	Absolute value.
$(\cdot)^{\mathrm{T}}$	Transpose operator.
$\|\mathcal{X}\|_1$	$\sum_{ijk}\|\mathcal{X}_{ijk}\|$.
x	Vector.
$\mathcal{X}$	Tensor.
$\|\mathbf{x}\|_2$	$\sqrt{\sum_i x_i^2}$.
$\odot$	Hadamard product.
$\|\mathcal{X}\|_{\mathrm{F}}$	$\sqrt{\sum_{ijk}\mathcal{X}_{ijk}^2}$.

Received 7 June 2025; revised 9 August 2025; accepted 7 September 2025. Date of publication 10 September 2025; date of current version 30 September 2025. *(Corresponding author: Qin Shu.)*

Maoyuan Feng and Qin Shu are with the College of Electrical Engineering, Sichuan University, Chengdu 610065, China (e-mail: 1170475472@qq.com; shuqin@scu.edu.cn).

Yunxiu Yang is with South-West Institute of Technical Physics, Chengdu 610041, China.

Yi Gan is with the 10th Research Institute, China Electronic Technology Group Corporation, Chengdu 610036, China.

Xiaoguang Shao is with Beijing Institute of Remote Sensing Equipment, Beijing 100854, China.

Digital Object Identifier 10.1109/TGRS.2025.3608242

I. INTRODUCTION

HYPERSPECTRAL image (HSI) captures spatially resolved spectral signatures across hundreds of contiguous narrow spectral bands [1], thereby providing ultrahigh spectral resolution capable of discerning subtle material-specific absorption and reflectance features. This unique capability enables precise characterization of target compositions, making HSI indispensable for applications such as land cover classification, environmental change monitoring, precision agriculture, and military surveillance [2]. Within the domain of hyperspectral image analysis, target detection has emerged as a critical research frontier, driven by advances in remote sensing technologies and computational methods [3]. Depending on the availability of prior spectral knowledge, hyperspectral target detection frameworks are broadly categorized into two paradigms: matched target detection [4], which requires prior spectral signatures (e.g., spectral libraries) to identify targets; and anomaly detection [5], which identifies statistically significant deviations from background distribution without prior spectral or spatial information. This article focuses on hyperspectral anomaly detection (HAD), a pivotal technique for uncovering spatially rare or unexpected targets (e.g., camouflaged vehicles, mineral deposits, or pollutant plumes) in complex scenes. By distinguishing anomalies based on their spectral–spatial incongruity with surroundings, HAD provides transformative solutions for diverse fields. Over the past two decades, extensive research efforts have been dedicated to addressing HAD challenges. We systematically review these methodologies in the following, analyze their theoretical foundations, and discuss their practical limitations.

A. Model-Driven-Based Methods

1) Distribution Assumption-Based Methods: The first category method relies on statistical assumptions, which generally assume that the background pixels adhere to a specific distribution, such as a Gaussian mixture model, while the anomaly

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

pixels do not conform to the same distribution. A widely recognized technique in this category is the Reed–Xiaoli (RX) method, proposed in [6]. The RX detector converts anomaly detection into the calculation problem of Mahalanobis distance through statistical modeling and hypothesis testing. It is a classic method for HAD. The key to its successful detection lies in the accuracy of the background statistical model. However, real-world HSI scenes often exhibit spatially heterogeneous backgrounds (e.g., varying terrain types or illumination conditions), rendering global statistical assumptions invalid. To address this limitation, the localized variants of RX were developed in [7], which uses an inner and outer double window: the inner window estimates the local background and the outer window eliminates potential abnormal interference and improves the robustness of background estimation. Furthermore, to solve the problem of nonlinear background distribution, a kernel-RX (KRX) was proposed in [8] by mapping the data into a high-dimensional feature space through kernel functions (such as Gaussian kernels). Besides, some other improved RX-based methods have been proposed, such as the cluster KRX in [9], the weighted-RX in [10], and so on.

2) Matrix Decomposition or Representation-Based Methods: With the development of matrix theory, some matrix decomposition techniques have also been introduced to help deal with the HAD task. The most common one is based on low-rank and sparse matrix decomposition (LRSMD) [11], [12], [13]. This type of algorithm straightens the hyperspectral cube into a matrix along the spectral dimension, decomposes the straightened matrix into a background matrix with low-rank properties and an anomaly matrix with sparse properties, and designs various low-rank and sparse constraints for the background and the anomaly matrices to achieve the separation. Besides, with the rapid development of signal processing technology, some representation-based methods were also proposed for HAD. Specifically, they are sparse-representation-based (SR) methods in [14] and [15], collaboration-representation-based (CR) methods in [16] and [17], and low-rank and sparse representation-based (LRSR) methods in [18], [19], [20], [21], [22], [23], and [24]. The distinctions among these three representation paradigms primarily stem from their unique dictionary construction philosophies and differential constraints imposed on the coefficient matrices. Specifically, the SR-based methods emphasize the completeness of the dictionary and the sparsity of the bases, the CR-based methods emphasize the cooperation of the dictionary basis, and the LRR-based methods emphasize the low rankness (may add the sparsity) of the coefficient matrix.

3) Tensor Decomposition or Representation-Based Methods: According to its imaging characteristics, HSI can be regarded as a 3-D tensor, denoted as $\mathcal{Y} \in \mathbb{R}^{b\times h\times w}$, where w, h, and b represent the width, height, and band of the HSI, respectively. However, the above matrix-based methods flatten 3-D HSI data into a matrix, only utilizing the spectral dimension information and ignoring the spatial neighborhood correlation, which may decrease the detection performance. Thanks to the development of data tensor processing technology, the multidimensional spatial–spectral data characteristics of HSI can be well exploited after adopting a tensor to represent HSI. In recent years, many scholars have introduced tensor processing technology into HAD, aiming to fully exploit the various potential characteristics of background and anomalies in the spatial–spectral dimension of hyperspectral and then design a reasonable background anomaly separation model. In tensor-based methods, the low rankness and piecewise smoothness of the background tensor, the group sparsity of the anomaly tensor are the key characteristics to separate the background and anomaly. In the early studies, scholars only used these two properties one-sidedly to design various anomaly detectors, such as tensor-completion-based method in [25], the tensor-decomposition-based methods in [26] and [27], and the tensor-representation-based methods in [28], [29], [30], and [31]. However, incomplete utilization of features may lead to anomaly detection performance degradation. Although the tensor-based methods proposed in [32], [33], and [34] take into account the low rankness and piecewise smoothness of the background, they overlook the fact that these two properties are present in the 3-D of the background. Considering this, some improved tensor-based methods were proposed in [35], [36], [37], and [38], which aim to jointly utilize the low rankness and piecewise smoothness of the background in 3-D.

B. Data-Driven-Based Methods

With the rapid development of deep learning technology, data-driven deep learning methods have achieved excellent performance in HAD. This type of method basically assumes that the background can be well reconstructed by the network, while the anomaly cannot be reconstructed. Therefore, most scholars regard the final reconstruction error of the network as an anomaly. Among them, the more famous one is the autoencoder-based (AE) method [39], [40], [41]. Some improved AE-based methods by introducing regularization on the latent feature to better separate the background and anomaly were proposed in [39], [42], and [43]. Wang et al. [44] proposed an end-to-end fully convolution network to construct the background from random white noise. A series of methods based on blind spot networks provides a new solution perspective for HAD and achieves good detection performance [45], [46], [47]. Another famous data-driven method is the GAN-based methods in [48], [49], and [50], which models the background distribution through an adversarial learning mechanism and uses the game relationship between the generator and the discriminator to identify anomalies. Besides, some transformer-based methods [40], [51], [52], [53], [54], [55] have also been proposed to deal with HAD tasks since the transformer can efficiently process long-distance dependencies and capture global information, due to the attention mechanism. Li et al. [51] proposed a random-masked transformer-based method for HAD, expected to build a unified anomaly detection model that only needs to be trained once. Young et al. [55] proposed a transformer-guided fractional attention in abundance domains. In [54], the author proposed a local and global transformer-based method to fully exploit the local and global information of the HSI. To comprehensively exploit the advantages of the local and global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

feature extraction of CNN and Transformer, some CNN-Transformer hybrid networks have been proposed for the HAD task [56], [57]. Recently, diffusion models have been applied in many image tasks, such as reconstructing high dynamic range images [58], [59], and so on. Also, some diffusion-based methods have been proposed for HAD in [60] and [61] to improve the anomaly detection performance.

C. Model-Data-Driven-Based Methods

In recent years, some methods have been proposed to utilize the advantages of data-driven and model-driven methods to fully exploit the characteristics of the background to guide network structure design and training. Those existing data-model-driven methods can be roughly divided into three categories: the first one is deep-unfolding-based methods proposed in [62], [63], [64], and [65], where the network architecture is constructed according to the mathematical expressions clearly defined in the model-based methods and the learnable parameters in the network are the regularization parameters or some variables in the equation. This type of method intends to solve the problem of selecting regularization parameters in model-based methods; the second category is the plug-and-play methods proposed in [23] and [66], which introduce a deep network trained on other datasets and impose constraints on certain variables in the mathematical expression of the model. This type of method intends to introduce additional deep priors for the model; the third is to impose mathematical constraints on the network output [67], [68], [69], [70], [71]. Compared with the first two types of methods, this type of method is more flexible. The network structure can be flexible and changeable, and the network parameters can be learned in the direction specified by different model-based regularizations. Although many model-data-driven methods have been proposed for HAD, the existing methods also have some disadvantages. For the deep-unfolding-based methods, the performance of those methods is affected by the inherent paradigm and cannot break through the shackles of the model itself; for the plug-and-play-based methods, the performance of those methods is affected by the model itself on the one hand and is also affected by the trained network on the other hand, after all, the network parameters are fixed; for those methods where the model-based regularization designed on the network output, most existing methods do not fully consider the multidimensional data characteristic of the background (which also exiting in the front type methods). Although the method in [71] takes the properties of the background in 3-D into account, it is still a bit difficult to learn different features simultaneously using a single deep transformation stage. Besides, imposing model-based constraints on the network output is not as good as integrating such constraints into the network [72]. Moreover, due to the diversity of imaging scenes, anomalies typically exhibit multiscale characteristics. Some researchers already indicate that fully extracting the multiscale features of hyperspectral images and effectively fusing them [73], [74], [75], [76] is very important for accurate background reconstruction. Unfortunately, the existing model-data driven methods do not take into account the problem of the multiscale features extraction and fusion in the network, which may lead to poor performance when facing multiscale anomaly detection.

D. Motivation and Contributions

To make full use of the advantage of the data-model-driven methods, exploit the latent properties of the background in 3-D, and take the multiscale of the anomaly into account at the same time, we merge the tensor decomposition framework into the network and proposed a two-stage deep transformation network with latent features guidance (TDTLFG) for HAD, where the tensor low-rank matrix factorization (LRMF) is merged into the network after the multiscale feature extraction (MSFE) transformation stage, and the total variation (TV) regularization is designed on the output after the multiscale feature fusion (MSFF) transformation stage. The proposed method combines the data characteristics guided regularization and the MSFE and fusion. On the one hand, it can fully utilize the potential data characteristics of the background tensor to guide the feature extraction and fusion (the MSFE guided by the low-rank property and the multilevel feature fusion guided by the piecewise-smooth property) to guide the network training with purposeful and directional. On the other hand, the proposed method utilizes the latent property of the background under different deep transformation domains instead of exploiting all properties under one image domain, realizing the decoupling of different properties. In summary, the deep transformations at different stages and the different background latent properties constraints complement each other and benefit from each other and work together to achieve better background reconstruction and anomaly detection results. The main contributions of this article are summarized as follows.

1) A data-model combined-driven MSFE and fusion network with latent data characteristic constraints is proposed for HAD, where the data characteristics are exploited in different deep transform domains to design a reasonable mathematical constraint to guide the network training, the rationality of the network design and the purpose and direction of the learning process are guaranteed under the guidance.
2) We propose a strategy to exploit the background data characteristics in the hierarchical deep transform domain. The tensor matrix factorization is embedded into the network as a low-rank constraint module after the first deep transformation stage, and the TV regularization is designed on the output of the second deep transformation stage as the piecewise smoothness constraint. Through different deep transformations, achieving background potential features, hierarchical extraction, and decoupled utilization.
3) To facilitate background reconstruction, MSFE and multilevel feature fusion modules are introduced. The former extracts multiscale features in the first transform stage, which is beneficial for scene content understanding, while the latter interactively fuses the multiscale features in the final transform stage, which is beneficial for background refinement.

The remainder of this article is organized as follows. Several definitions, notations, and related works are introduced in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

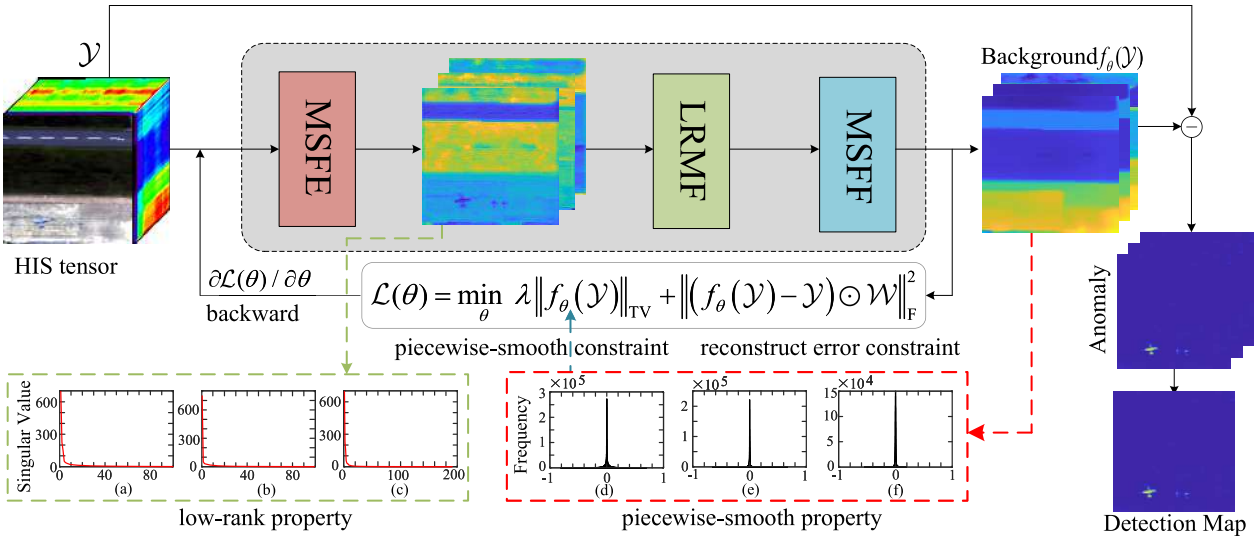


Fig. 1. Flowchart of the proposed method. (a)–(c) Eigenvalues of the unfolding matrices of the feature tensor $f_{\theta_{\mathrm{msfe}}}(\mathcal{Y})$ after the first MSFE transformed stage. (d)–(f) Histogram of the element of $\mathbf{D}_1(f_\theta(\mathcal{Y}))_{(1)}$, $\mathbf{D}_2(f_\theta(\mathcal{Y}))_{(2)}$, and $\mathbf{D}_3(f_\theta(\mathcal{Y}))_{(3)}$, respectively. (a)–(c) Low rankness does exist in the 3-D of the tensor $f_{\theta_{\mathrm{msfe}}}(\mathcal{Y})$. (d)–(f) Piecewise-smoothness does exist in the 3-D of the tensor $f_\theta(\mathcal{Y})$. Thus, the LRMF module designed on the first transformed stage and the TV regularization designed on the network output are interpretable.

Section II. The proposed method and the solution scheme are given in Section III. A series of simulation results conducted on several real datasets is given in Section IV. Finally, a brief summary is given in Section V.

II. Preliminaries

A. Tensor Nation

To establish the mathematical foundation for subsequent analysis, we first summarize key notations and their interpretations in Nomenclature, which will be used in the following. Then, we introduced some definitions and operations of tensor.

Definition 1 (Mode-k Unfolding): The mode-k unfolding $\mathbf{X}_{(k)}$ of an Nth order tensor $\mathcal{X} \in \mathbb{X}^{I_1\times I_2\times\cdots\times I_N}$ is an $I_k \times \prod_{n\neq k} I_n$ matrix, which satisfies that $\mathbf{X}_{(k)}(i_k, t)$ is mapped by $\mathcal{X}(i_1, i_2, \ldots, i_N)$, where $t = 1 + \sum_{(n=1,n\neq k)}^{N}(i_n - 1) * J_n$ with $J_n = \sum_{r=1,r\neq k}^{n-1} n_r$. The corresponding operator and inverse operator are denoted as $\mathbf{X}_{(k)} = \mathrm{unfold}_k(\mathcal{X})$ and $\mathcal{X} = \mathrm{fold}_k(\mathbf{X}_{(k)})$, respectively.

III. Proposed Method

As we mentioned above, although the combination of model-driven and data-driven methods has many advantages, the existing model and data-driven methods still have much to improve. Based on this, we propose a tensor low-rank matrix decomposition guidance and physical feature constraint two-stage transformation networks for HAD from the perspectives of model-driven constraints and data-driven MSFE and fusion. The flowchart of the proposed method is shown in Fig. 1, we will introduce the details of the proposed method in the following.

A. Framework of the Proposed Method

1) MSFE Module: HSI often presents uneven background distribution and diversity of anomaly scales as the scene changes. On the one hand, anomaly can be as small as a single pixel (such as a metal fragment) or as large as a continuous area (such as a contaminant diffusion area), on the other hand, hyperspectral scenes usually contain a variety of ground objects (such as vegetation, soil, and water bodies), and their spatial distribution characteristics vary significantly. Therefore, it is difficult to effectively extract background features and achieve accurate background reconstruction using a single-scale feature extraction module. To improve the robustness and adaptability of the detection model, we adopted an MSFE module to capture different-scale features; the whole structure of the proposed MSFE is displayed in Fig. 2. As we can see from Fig. 2, the proposed architecture incorporates three parallel processing branches comprising depth-wise separable convolutions with strategically configured kernel dimensions (3×3, 5×5, and 7×7) to systematically capture (a) fine-grained local patterns, (b) intermediate contextual cues, and (c) high-level semantic abstractions, establishing a multiscale feature hierarchy through differential receptive field expansion, which can be mathematically represented as

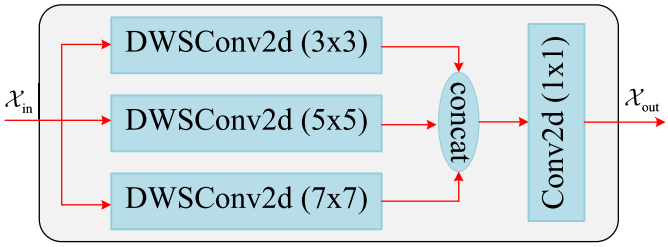


Fig. 2. Architecture of the MSFE module.

$$\mathcal{F}_i^e = \mathrm{DWSConv2d}_{ks_i}(\mathcal{Y}) \in \mathbb{R}^{b\times h\times w}, \quad (i = 1, 2, 3) \tag{1}$$

where $\mathrm{DWSConv2d}_{ks_i}(\cdot)$ denotes the depth-wise separate convolution layer with kernel size being ks_i (here, $ks_1 = 3$, $ks_2 = 5$, and $ks_3 = 7$). The depth-wise separable convolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

strategically decouples standard convolution into two sequential operations: channel-specific spatial filtering via depthwise convolution (implemented by setting the "group" equal to the input channel dimension) and cross-channel feature fusion through pointwise convolution. This decomposition achieves dramatic parameter reduction while preserving representational capacity, enabling efficient computation without compromising feature discriminability. After extracting different scale features, in order to simplify the process, we use a simple channel splicing and fusion method to splice features of different scales along the channel dimension and then use a convolution layer to fuse and adjust the number of channels. The details are as follows:

$$\mathcal{F}^{ef} = \text{Conv2d}_{ks_f}\left(\text{Cat}\left(\left[\mathcal{F}_1^e, \mathcal{F}_2^e, \mathcal{F}_3^e\right]\right)\right) \in \mathbb{R}^{b\times h\times w} \tag{2}$$

where $\text{Cat}(\cdot)$ denotes the concatenation and $\text{Conv2d}_{ks_f}(\cdot)$ denotes the "Conv2D" convolution layer with kernel size bing ks_f (here, we set $ks_f = 1$).

2) LRMF Low-Rank Constraint Module: In fact, the low rankness was observed across all dimensions of the background tensor. This property has been widely used in various HAD methods and it has been proven that this property can improve the detection performance to a certain extent. In our previous work [71], we also revealed that the background feature tensor after deep transformation has low rankness in 3-D. Then, the background feature tensor must also have low rankness in 3-D after the MSFE module proposed in this article. We perform eigenvalue analysis on each dimension of the feature tensor after MSFE, and the eigenvalue of each dimension is shown in Fig. 1. As we can see from Fig. 1, the feature tensor obtained by the MSFE module does have a strong low-rank property in each dimension, which is consistent with the strong low-rank property in each dimension of the original background. To holistically exploit the multidimensional low rankness of the background after the first deep transformation stage, we architect a triple-parallel tensor decomposition framework through LRMF behind the first MSFE deep transformation stage. Specifically, the spectral dimension branch operates through

$$\begin{cases} \mathbf{F}_1 = \text{Unfold}_1\left(\mathcal{F}^{ef}\right) \in \mathbb{R}^{b\times wh} \\ \mathbf{U}_b = f_{\mathbf{U}_b}\left(\mathbf{F}_{(1)}\right) \in \mathbb{R}^{b\times r_b} \\ \mathbf{V}_b = f_{\mathbf{V}_b}\left(\mathbf{F}_{(1)}^{\mathrm{T}}\right) \in \mathbb{R}^{wh\times r_b} \\ \mathbf{F}_{lr,b} = \mathbf{U}_b\mathbf{V}_b^{\mathrm{T}} \in \mathbb{R}^{b\times wh} \\ \mathcal{F}_{lr,b} = \text{Fold}_{(1)}\left(\mathbf{F}_{lr,b}\right) \in \mathbb{R}^{b\times h\times w} \end{cases} \tag{3}$$

where $f_{\mathbf{U}_b}(\cdot)$ and $f_{\mathbf{V}_b}(\cdot)$ represent two linear-layer modules (denoted as "Linear" in Fig. 3) that perform low-rank matrix decomposition to generate the low-rank matrix factors in the band dimension. As shown in Fig. 3, the first layer in $f_{\mathbf{U}_b}(\cdot)$ is a linear layer with input shape being wh and output shape being r_b. The structure of module $f_{\mathbf{V}_b}(\cdot)$ is the same as that of the module $f_{\mathbf{U}_b}(\cdot)$. The only difference is that the input shape of the first linear layer in module $f_{\mathbf{V}_b}(\cdot)$ is b. Similarly, for the height dimension, we have $\mathbf{U}_h = f_{\mathbf{U}_h}\left(\mathbf{F}_{(2)}\right) \in \mathbb{R}^{h\times r_h}$, $\mathbf{V}_h = f_{\mathbf{V}_h}(\mathbf{F}_{(2)}^{\mathrm{T}}) \in \mathbb{R}^{wb\times r_h}$, $\mathbf{F}_{lr,h} = \mathbf{U}_h\mathbf{V}_h^{\mathrm{T}} \in \mathbb{R}^{h\times wb}$, and $\mathcal{F}_{lr,h} = \text{Fold}_{(2)}\left(\mathbf{F}_{lr,h}\right) \in \mathbb{R}^{b\times h\times w}$, where the structure of modules $f_{\mathbf{U}_h}(\cdot)$ and $f_{\mathbf{V}_h}(\cdot)$ are the same as that of module $f_{\mathbf{U}_b}(\cdot)$ with the only difference is that the input and output shape of the module $f_{\mathbf{U}_h}(\cdot)$ are wb and r_h, respectively, and the input and output shape of the module $f_{\mathbf{V}_h}(\cdot)$ are h and r_h, respectively. For the weight dimension, we have $\mathbf{U}_w = f_{\mathbf{U}_w}\left(\mathbf{F}_{(3)}\right) \in \mathcal{R}^{w\times r_w}$, $\mathbf{V}_w = f_{\mathbf{V}_w}\left(\mathbf{F}_{(3)}^{\mathrm{T}}\right) \in \mathbb{R}^{hb\times r_w}$, $\mathbf{F}_{lr,w} = \mathbf{U}_w\mathbf{V}_w^{\mathrm{T}} \in \mathbb{R}^{w\times hb}$, and $\mathcal{F}_{lr,w} = \text{Fold}_{(3)}\left(\mathbf{F}_{lr,w}\right) \in \mathbb{R}^{b\times h\times w}$, where modules $f_{\mathbf{U}_w}(\cdot)$ and $f_{\mathbf{V}_w}(\cdot)$ have the same structure as module $f_{\mathbf{U}_b}(\cdot)$ with the input and output shape of the model $f_{\mathbf{U}_w}(\cdot)$ are hb and r_w, and the input and output shape of the model $f_{\mathbf{V}_w}(\cdot)$ are w and r_w, respectively. After three parallel independent LRMF branches, the fusion feature low-rank tensor is calculated via

$$\mathcal{F}_{lr}^{fs} = \frac{\mathcal{F}_{lr,b} + \mathcal{F}_{lr,h} + \mathcal{F}_{lr,w}}{3}. \tag{4}$$

The low rankness of the background feature tensor in 3-D is effectively exploited to guide network training through the LRMF module. The benefit of the LRMF module is that, on the one hand, it uses a model-driven approach to design the network structure, making the network interpretable; on the other hand, the module can be used as a soft constraint for the MSFE module, using background characteristics to guide the front-end network learning, making the first-stage deep transformation purposeful and directional.

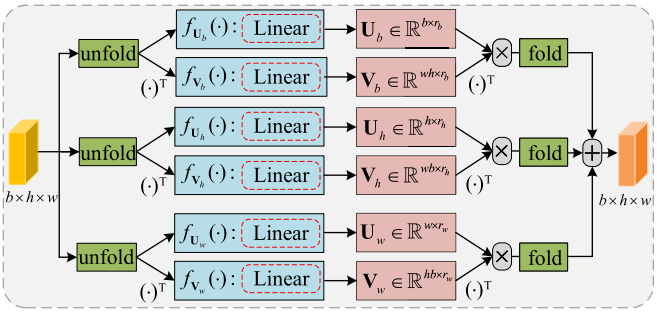


Fig. 3. Architecture of the LRMF module.

3) MSFF Module: After adopting the LRMF module to exploit the low rankness of the background in 3-D, we introduced a Unet with MSFF (which is inspired by [77]), called MSFF-Unet module, to achieve information exchange between cross-scale features. The structure of this network is shown in Fig. 4, which consists of a standard Unet with an MSFF model. The Unet can be divided into the downsample, the upsample, and the final tail streams. In the downsample stream, the network consists of three downsample blocks; the detail of the downsample block is displayed in Fig. 5. As we can see from Fig. 5, each downsample block consists of three "Conv2d" layers, and each "Conv2d" layer is followed by a "BatchNorm2d" and a "LeakyReLU" layer. In all downsample blocks, except for the first "Conv2d" of the first downsample block whose input channel dimension is b, the number of the input and output channels of all other "Conv2d" is 32. In the first "Conv2d" layer, the kernel size and the stride are 1. In the second "Conv2d" layer, the kernel size is 3, the stride is 2, and the padding is 1. In the third "Conv2d" layer, the kernel size is 3, the striding is 1, and the padding is 1. After each downsample block, the spatial dimension of the feature map is halved, and the channel dimension remains unchanged. Thus,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

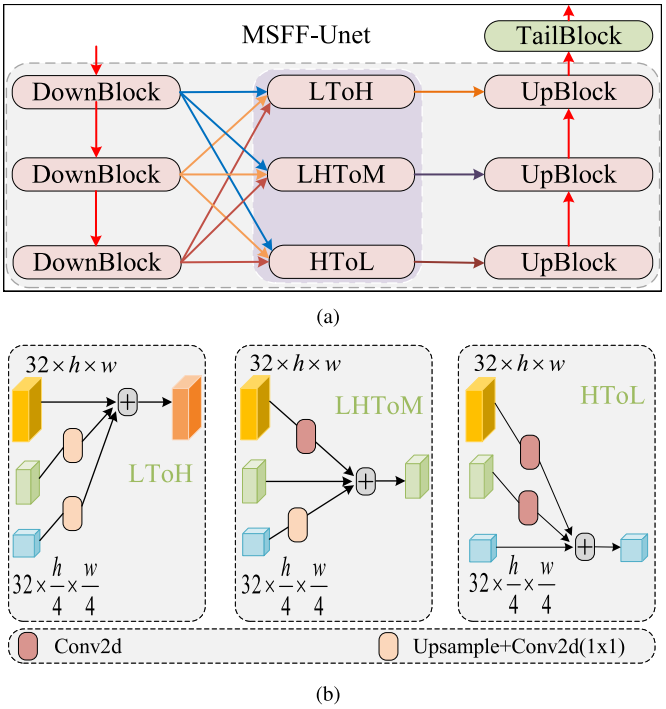


Fig. 4. Structure of the Unet with MSFF module. (a) Total MSFF-Unet structure. (b) Detail of the MSFF module.

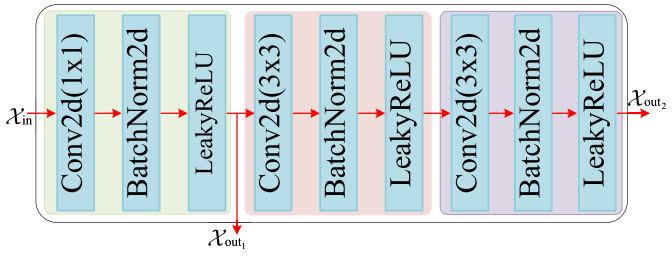


Fig. 5. Architecture of the downsample block, where $\mathcal{X}_{\text{in}}$ and $\mathcal{X}_{\text{out}_1}/\mathcal{X}_{\text{out}_2}$ represent the input and outputs of the downsample block, respectively.

the feature tensors after the first, second, and third downsample modules are expressed as

$$\begin{cases} \mathcal{D}_1 = f_{\text{ds}}^1(\mathcal{F}_{lr}^{fs}) \in \mathbb{R}^{32\times h\times w} \\ \mathcal{D}_2 = f_{\text{ds}}^2(\mathcal{D}_1) \in \mathbb{R}^{32\times \frac{h}{2}\times \frac{w}{2}} \\ \mathcal{D}_3 = f_{\text{ds}}^3(\mathcal{D}_2) \in \mathbb{R}^{32\times \frac{h}{4}\times \frac{w}{4}} \end{cases} \tag{5}$$

where $f_{\text{ds}}^n(\cdot), n = (1, 2, 3)$ denotes the "DownBlock" in Fig. 4. In the traditional Unet, the different resolution downsample feature map is spliced with the corresponding upsample feature map and enters the upsample module. Although different resolution features interact across resolutions after fusion, they cannot interact directly, which may lead to insufficient utilization and interaction of different resolution features. Thus, inspired by [77], we proposed an MSFF module that enables different resolution features to interact directly. The detail of the MSFF module is displayed in Fig. 4(b). As we can see from Fig. 4(b), the low-resolution feature map becomes a high-resolution feature after passing an upsample module, where the upsample module consists of an "Upsample" layer and a "Conv2d" layer with kernel size and stride being 1;

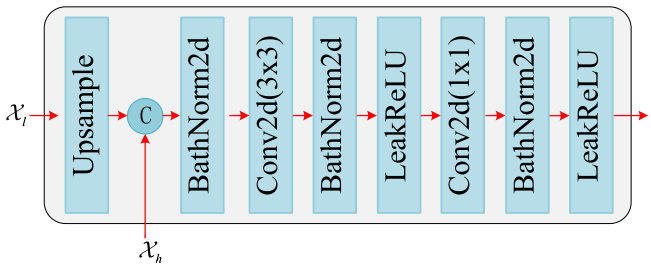


Fig. 6. Architecture of the upsample block, where $\mathcal{X}_h$ and $\mathcal{X}_l$ represents the high-resolution and low-resolution feature maps, respectively.

for the high-resolution feature, it becomes a low-resolution feature through a downsample "Conv2d" layer with groups be the same shape as the input shape. As we can see from Fig. 4(b), we get three different resolution feature maps after three downsample blocks, denoted as high-, medium-, and low-resolution feature maps. Thus, we design three types of fusion modules: they are low- and medium-resolution feature maps fuse with high-resolution feature map; high- and low-resolution feature maps fuse with medium-resolution feature map; and high- and medium-resolution feature maps fuse with low-resolution feature map. We denote the corresponding modules as "LToH," "LHToM," and "HToL." The mathematical expressions of the three modules are expressed as

$$\begin{cases} \text{LToH: } \mathcal{F}_f^h = \mathcal{D}_1 + f_{sf:2}^{\text{up:2}\to 1}(\mathcal{D}_2) + f_{sf:4}^{\text{up:3}\to 1}(\mathcal{D}_3) \\ \text{LHToM: } \mathcal{F}_f^m = \mathcal{D}_2 + f_{sd:2;ks:3;pd:1}^{\text{dw:1}\to 2}(\mathcal{D}_1) + f_{sf:2}^{\text{up:3}\to 2}(\mathcal{D}_3) \\ \text{HToL: } \mathcal{F}_f^l = \mathcal{D}_3 + f_{sd:4,ks:5,pd:2}^{\text{dw:1}\to 3}(\mathcal{D}_1) + f_{sd:2,ks:3,pd:1}^{\text{dw:2}\to 3}(\mathcal{D}_2) \end{cases} \tag{6}$$

where $f_{sf:2}^{\text{up:2}\to 1}(\cdot)$ and $f_{sf:4}^{\text{up:3}\to 1}(\cdot)$ denote the upsample modules in LToH, in which $sf : 2$ represents the scale factor (sf) is 2 and $f_{sf:2}^{\text{up:3}\to 2}(\cdot)$ represents the upsample module in LHToM; $f_{sd:4,ks:5,pd:2}^{\text{dw:1}\to 3}$ and $f_{sd:2,ks:3,pd:1}^{\text{dw:2}\to 3}$ represent the downsample "Con2d" layer in HToL, where the $sd : 4$, $ks : 5$, and $pd : 2$ represent the stride is 4, the kernel size is 5, and the padding is 2, respectively; $f_{sd:2,ks:3,pd:1}^{\text{dw:1}\to 2}$ represents the downsample module in HLToM; and $\mathcal{F}_f^h$, $\mathcal{F}_f^m$, and $\mathcal{F}_f^l$ denote the fusion high-, medium-, and low-resolution feature maps, respectively.

After getting the three different resolution fusion features, the process is the same as the traditional Unet, where those features are fed into the upsample stream. The upsample stream consists of three upsample blocks (denoted as "UpBlock" in Fig. 4); the details of each block are displayed in Fig. 6, where the input and the output channel of the "Conv2d" layer are 64 and 32, respectively; the kernel size of the first and second "Conv2d" layers are 3 and 1, respectively; and the stride of all "Conv2d" layers are 1. Finally, the output of the last upsample block is fed into the tail block, which consists of a "Conv2d" and a "Sigmoid" layer. The input and output channels of the "Conv2d" are 64 and b, respectively, and the kernel size and stride of this "Conv2d" layer are 1.

4) Loss Function: After the MSFF module, on the one hand, to achieve the step-by-step utilization of the low-rank and piecewise-smooth properties of the background in different deep transformation domains, we consider designing a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

3-DTV constraint on the output of the network to exploit the inherent piecewise-smooth property of the background in 3-D [actually, the piecewise-smooth property does exist in the 3-D of the network output tensor (as shown in Fig. 1)]. On the other hand, considering the basic principles of background reconstruction, refer to [44] and [67], we design a weighted reconstruction error constraint on the output of the network. Therefore, the final network cost function is designed as

$$\mathcal{L}(\theta) = \|\mathcal{W} \odot (\mathcal{Y} - f_\theta(\mathcal{Y}))\|_{\mathrm{F}}^2 + \lambda \|f_\theta(\mathcal{Y})\|_{\mathrm{TV}} \tag{7}$$

where $\mathcal{W} \in \mathbb{R}^{b\times h\times w}$ is the weight coefficient, which is calculated according to [67]; $f_\theta(\cdot)$ denotes the network proposed in this article; θ is the network parameter needed to be learned; and λ is a nonnegative weight coefficient to trade off the different regularization parts; $\|\cdot\|_{\mathrm{TV}}$ represents the 3DTV in this article. Specially, for a three-order tensor $\mathcal{T}$, the 3DTV is expressed in detail as $\|\mathcal{T}\|_{\mathrm{TV}} = \sum_{n=1}^{3} \|\nabla_n(\mathcal{T})\|_1$, where $\nabla_n(\mathcal{T})(n = 1, 2, 3)$ is calculated via

$$\begin{cases} \nabla_1(\mathcal{T})(i,j,k) = \mathcal{T}(i+1,j,k) - \mathcal{T}(i,j,k) \\ \nabla_2(\mathcal{T})(i,j,k) = \mathcal{T}(i,j+1,k) - \mathcal{T}(i,j,k) \\ \nabla_3(\mathcal{T})(i,j,k) = \mathcal{T}(i,j,k+1) - \mathcal{T}(i,j,k). \end{cases} \tag{8}$$

Here, we adopted the alternating direction method of multipliers (ADMM) to solve the parameters involved in (7). After introducing three auxiliary parameters to simplify the process, the loss function can be rewritten as

$$\begin{cases} \arg\min\limits_{\theta,\mathcal{Z}_n} \|\mathcal{W} \odot (\mathcal{Y} - f_\theta(\mathcal{Y}))\|_{\mathrm{F}}^2 + \lambda \sum\limits_{n=1}^{3} \|\mathcal{Z}_n\|_1 \\ \text{s.t. } \nabla_n(f_\theta(\mathcal{Y})) = \mathcal{Z}_n, \quad (n = 1, 2, 3). \end{cases} \tag{9}$$

Thus, the augmented Lagrangian function of (9) can be written as

$$\begin{aligned} &\mathcal{L}(\theta, \mathcal{Z}_n, \mathcal{P}_n) \\ &= \|\mathcal{W} \odot (\mathcal{Y} - f_\theta(\mathcal{Y}))\|_{\mathrm{F}}^2 + \lambda \sum_{n=1}^{3} \|\mathcal{Z}_n\|_1 \\ &\quad + \frac{\mu}{2} \sum_{n=1}^{3} \|\nabla_n(f_\theta(\mathcal{Y})) - \mathcal{Z}_n\|_{\mathrm{F}}^2 + \langle \mathcal{P}_n, \nabla_n(f_\theta(\mathcal{Y})) - \mathcal{Z}_n \rangle \end{aligned} \tag{10}$$

where μ is a nonnegative tradeoff parameter and $\mathcal{P}_n$ is the Lagrangian multiplier. Then, (10) can be solved through the following subproblems.

1) Update θ, and the loss function with respect to parameter θ can be written as

$$\begin{aligned} \mathcal{L}(\theta) = \arg\min_{\theta} &\|\mathcal{W} \odot (\mathcal{Y} - f_\theta(\mathcal{Y}))\|_{\mathrm{F}}^2 \\ &+ \frac{\mu^t}{2} \left\| \nabla_n(f_\theta(\mathcal{Y})) - \mathcal{Z}_n + \frac{\mathcal{P}_n^t}{\mu^t} \right\|_{\mathrm{F}}^2 \end{aligned} \tag{11}$$

where t represents the number of iterations. The parameter θ in (11) can be optimized using gradient methods, and we adopt the "Adam" optimizer for this purpose in our implementation.

2) Update $\mathcal{Z}_n(n = 1, 2, 3)$, and the loss function with respect to parameter $\mathcal{Z}_n$ can be written as

$$\begin{aligned} \mathcal{L}(\mathcal{Z}_n) = \arg\min_{\mathcal{Z}_n} &\lambda \|\mathcal{Z}_n\|_1 \\ &+ \frac{\mu^t}{2} \left\| \nabla_n(f_{\theta_{t+1}}(\mathcal{Y})) - \mathcal{Z}_n + \frac{\mathcal{P}_n^t}{\mu^t} \right\|_{\mathrm{F}}^2. \end{aligned} \tag{12}$$

Denote $\bar{\mathcal{Z}}_n = \nabla_n(f_{\theta_{t+1}}(\mathcal{Y})) + \mathcal{P}_n^t/\mu^t$. Thus, (12) can be solved by

$$\mathcal{Z}_n^{t+1}(i_b, i_h, i_w) = \mathrm{soft}_1\left(\bar{\mathcal{Z}}_n(i_b, i_h, i_w), \frac{\lambda}{\mu^t}\right) \tag{13}$$

where $i_w \in (1, 2, \ldots, w)$, $i_h \in (1, 2, \ldots, h)$, $i_b \in (1, 2, \ldots, b)$, and $\mathrm{soft}_1(\cdot)$ is defined as

$$\mathrm{soft}(x, a) = \mathrm{sign}(x) \max(|x|, a) \tag{14}$$

where $\mathrm{sign}(\cdot)$ is the sign function.

3) *Update Multipliers:* The parameters μ and $\mathcal{P}_n$ can be updated via

$$\begin{cases} \mathcal{P}_n^{t+1} = \mathcal{P}_n^t + \mu^t \left(\nabla_n(f_{\theta_{t+1}}(\mathcal{Y})) - \mathcal{Z}_n^{t+1}\right) \\ \mu^{t+1} = \min\left(\rho\mu^t, \mu_{\max}\right) \end{cases} \tag{15}$$

where $\rho > 0$ is a nonnegative parameter and $\mu_{\max}$ is the up bound of μ. Similarly, we adopted the termination judgment condition in [67], that is,

$$\varsigma = \sum_{t=t'}^{t'+50} \left|\mathcal{L}(\theta_{t+1}) - \mathcal{L}(\theta_t)\right|. \tag{16}$$

When the iteration reaches the maximum number of iterations or the above convergence condition, the estimate of the anomaly tensor is recorded as $\tilde{\mathcal{A}} = \mathrm{reshape}(\mathcal{Y} - f_\theta(\mathcal{Y})) \in \mathbb{R}^{w\times h\times b}$. Finally, the anomaly map, denoted as $\tilde{\mathbf{A}} \in \mathbb{R}^{w\times h}$, is calculated by

$$\tilde{\mathbf{A}}(i_w, i_h) = \sum_{i_b=1}^{b} \left\|\tilde{\mathcal{A}}(i_w, i_h, i_b)\right\|^2 \tag{17}$$

and $\tilde{\mathbf{A}}$ is normalized to obtain the final anomaly detection map via

$$\bar{\tilde{\mathbf{A}}} = \frac{\tilde{\mathbf{A}} - \min(\tilde{\mathbf{A}})}{\max(\tilde{\mathbf{A}}) - \min(\tilde{\mathbf{A}})}. \tag{18}$$

5) Rank Initialization: In practice, we cannot know the true rank of the unfolding matrices $\mathbf{Y}_i$, $(i = 1, 2, 3)$. Thus, it is necessary to develop a useful approach to obtain the approximate rank of those matrices. The rank-decreasing method in [78] can be used to estimate r_h, r_w, and r_b. However, this method needs to estimate the three parameters through iteration. Each iteration still needs to perform SVD on unfolding matrices; the huge amount of calculation brought by SVD is inevitable when this process needs many iterations. To avoid computational complexity brought by multitimes SVD, we adopt a one-time initialization approach, which means that once the initialization is determined, the values of r_w, r_h, and r_b will no longer change during parameter iteration updates. We adopted a simple but efficient approach to estimate the three parameters. By performing EVD to obtain the eigenvalues of $\mathbf{Y}_{(i)}\mathbf{Y}_{(i)}^{\mathrm{T}}(i = 1, 2, 3)$ and then sort the corresponding eigenvalues, we can obtain $\eta_{i,1} > \eta_{i,2} > \cdots > \eta_{i,n_i}, (n_i = \{w, h, b\})$. Then, we count the number of eigenvalues, which satisfies $(\eta_{i,n_j})^{1/2} > (\eta_{i,1})^{1/2} * \rho_\eta, (1 \le n_j \le n_i)$ as the rank of the model-i unfolding matrix, where ρ_η is a constant, and we set it as 0.01 in this article.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

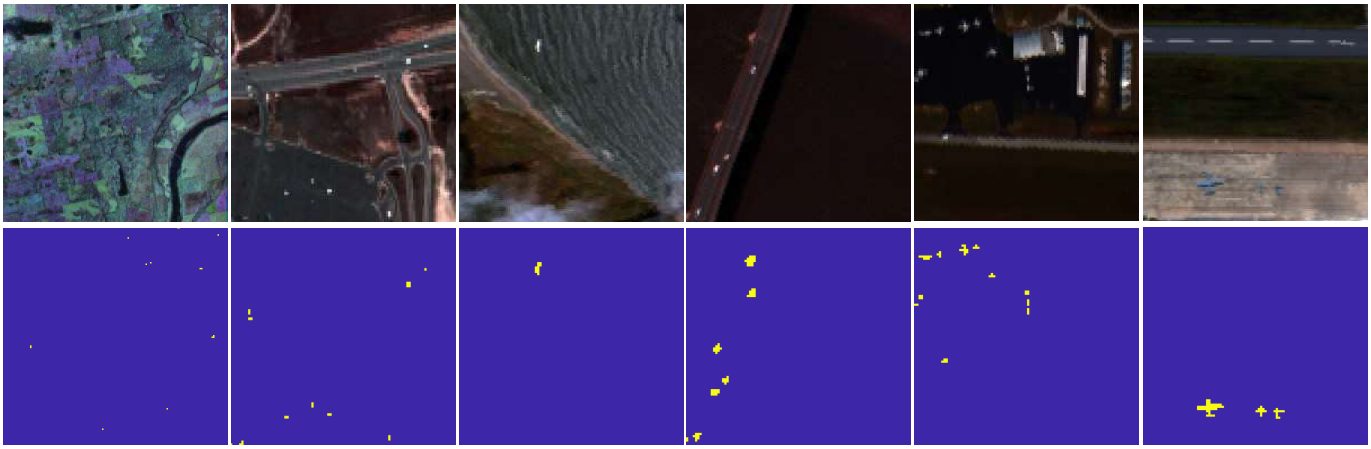


Fig. 7. Pseudocolor and anomaly ground truth of the experimental datasets. (a)–(f) Hyperion, HYDICE, beach-3, Pavia, urban-3, and airport-4 datasets, respectively.

Algorithm 1 Proposed TDTLFG Method

Input: Data tensor $\mathcal{Y}$, parameters λ, $[r_b, r_h, r_w]$, and $t_{\max}$.

Initialize: The variables $\mathcal{P}_n$ are initialized to zeros, $\rho = 1.02$, $\mu_0 = 0.5$, $\mu_{\max} = 10^9$, $\varsigma = 10^{-6}$.

While ($\varsigma > \varsigma_0$ and $t < t_{\max}$) **do**

1) Update θ via eq(11).
2) Update $\mathcal{Z}_n$ via eq(12).
3) Update multipliers via eq(15).
3) Update ς via eq(16).
4) $t = t + 1$.

End while

Output Obtain $\bar{\bar{\mathbf{A}}}$ according to (17) and (18).

The primary solution steps of the proposed method are given in Algorithm 1, where $t_{\max}$ is the maximum iteration, ε_0 is the set stopped iteration threshold, and μ_0 is the initialization value of μ.

IV. Simulation Results

This section presents the experimental evaluation of HAD performance across six hyperspectral datasets. All computational experiments were executed on a workstation equipped with an Intel Core i9-12900K processor (3.20 GHz), 64-GB RAM, and an NVIDIA RTX 3090 Ti GPU under a Windows 10 environment. In the following experiment, the deep-learning-based methods were implemented with PyTorch 1.13.0, while the model-based methods were programmed in MATLAB 2022a. The following section will provide the detailed experimental and analytical results that comprehensively demonstrate the superiority of the proposed algorithm through multiple evaluative dimensions.

A. Hyperspectral Datasets

In this article, six datasets are adopted to carry out the experiments. The first dataset, Hyperion, a common dataset for small-target detection, comprises 150×150 spatial pixels with 155 effective spectral bands. Here, ten anomalies, a total of 15 pixels, are taken as anomalies. The second dataset, HYDICE-urban (demoted as HYDICE), has 80×100 pixels and 175 effective bands. This dataset highlights the mixed-scale anomaly detection challenge, which is characterized by the coexistence of subpixel anomalies with small to medium-sized objects. The third dataset, abu-beach-3 (denoted as beach-3), comprises 100×100 pixels and 188 effective bands. This is a relatively simple medium-sized anomaly detection scenario, where a single object consisting of 11 pixels is considered an anomaly. The fourth dataset, Pavia Center, contains 100×100 pixels and 102 effective bands. This is a medium-to-large target detection scenario; six cars with 71 pixels are taken as anomalies in this scene. The fifth dataset, abu-urban-3 (simply denoted as urban-3), contains 100×100 pixels and 191 effective bands. This is a mixture-size anomaly detection scenario, about 7 small-size and 2 medium-size targets containing a total of 52 pixels are taken as anomalies. The sixth dataset, abu-airport-4 (simply denoted as airport-4), contains 100×100 pixels and 205 effective bands. This is a typical large-scale anomaly detection scenario, where three planes contain 60 pixels in total and are considered as anomalies. The Hyperion can be found in [45], and other datasets can be found in [79]. The pseudo-color images and anomaly ground-truth map for the six datasets are shown in Fig. 7.

B. Evaluation Indicators and Comparison Methods

The following conditions hold.

1) This article adopted five common evaluation indicators to measure the detection performance of different methods, which are the anomaly detection map, the 2-D receiver operating characteristic (ROC) curve, the area under the ROC curve (AUC) (i.e., the $\text{AUC}_{(\text{PD,PF})}$, $\text{AUC}_{(\text{PD},\tau)}\text{AUC}_{(\text{PF},\tau)}$, $\text{AUC}_{(\text{ODP})}$, and $\text{AUC}_{(\text{SNPR})}$ in [80]), the anomaly-background separation map, and the running time to evaluate the anomaly detection, background suppression, target background separation per-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

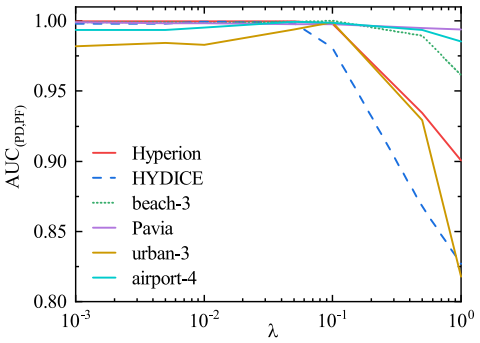


Fig. 8. $AUC_{(PD,PF)}$ values of the proposed method under different value of λ.

formance, and algorithm operation efficiency of different algorithms.

2) To demonstrate the superiority of the proposed method, we introduce ten comparison methods covering different principles for comparison, including one improved static-based method called LRX in [7]; two matrix-based methods called LSMAD in [12], and GTVLRR in [21]; two tensor-based methods called LARTVAD in [33] and PCA-TLRSR in [28]; two data-driven-based methods called TAEF in [40] and MSNet in [74]; and three data-model-driven-based methods call DeepLR in [67], LRR-Net-S in [63], and NLDPAD in [66]. The inner window size and the outer window size in the LRX are set to 3 and 5, respectively. The parameters r and k in LSMAD are chosen from the given sets in [12]. The parameters λ, β, and γ in GTVLRR are chosen from $\{0.005, 0.01, 0.1, 0.2, 0.3, 0.5, 1, 2, 3\}$. The parameters λ and β in LARTVAD are chosen from the set $\{0.01, 0.1, 1, 2, 3, 10\}$. Other parameters remain unchanged. The number of clusters in GTVLRR and LARTVAD is set to 15, and 20 pixels are selected to construct the atom in each cluster when building the background dictionary. The parameters λ and λ' in PCA-TLRSR are chosen from the set $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$. The parameters involved in DeepLR are chosen according to [67]. The parameter initialization in LRR-Net-S is the same as that introduced by Li et al. [63]. Similarly, the parameters involved in NLDPAD are chosen from the given range as introduced in Ren et al. [66].

C. Detection Performance Analysis via Parameters λ

In the proposed method, the value of the parameter λ has a certain impact on the detection performance. In this section, the detection performance of the proposed method on the experimental dataset under different λ values is analyzed to exploit the robustness of the proposed method to the parameter λ. In this article, the value of λ is chosen from the set $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1\}$. Fig. 8 shows the detection results of the proposed method under different lam values on the six experimental datasets (for the convenience of analysis, we selected the $AUC_{(PD,PF)}$ value as the measurement indicator of the detector), and the results are displayed in Fig. 8. Fig. 8 shows that the $AUC_{(PD,PF)}$ value increase when $\lambda < 0.1$, but the change is not obvious; the value of $AUC_{(PD,PF)}$ shows an intensified decreasing trend when $\lambda > 0.1$. The six datasets get the best $AUC_{PD,PF}$ values when λ set the values as 0.01, 0.01, 0.1, 0.01, 0.1, and 0.05. Therefore, we set λ as those values for those six datasets in the following simulations.

D. Experimental Results and Analysis

This section visually compares detection performance through experimental results obtained by different comparison methods across six commonly used datasets. Figs. 9–14 present anomaly detection maps obtained by all comparison methods across the whole experimental datasets. These visualizations show that the compared methods based on deep unfolding and plug-and-play are generally inferior to the proposed method and the method that designs model-based constraints on the network output in terms of both anomaly detection and background suppression, which largely indicates that the model itself constrains these two types of methods. Besides, the data-driven methods and the LRX demonstrate superior background suppression performance compared to the model-based methods, attributable to their effective utilization of nonlocal contextual information. However, the traditional deep-learning-based methods weaken the anomalies during background suppression, a limitation particularly pronounced in large-target scenarios, which can be seen from the airport-4 dataset in Fig. 14. In contrast, the proposed method achieves simultaneous background suppression and anomaly enhancement across varying target scales (which can bee see from the airport-4 dataset, where the proposed method has excellent anomaly detection performance while completely suppressing the bridge in the background) through comprehensive exploitation of the intrinsic background characteristics: global low-rank structure and piecewise-smooth property of the background in 3-D, where those properties do exist in the 3-D of the feature map tensor after the first and second deep transformation domain. This capability is further reinforced by embedding the low-rank constraint into the inner network structure and designing piecewise-smooth constraints on the output of the network to guide background reconstruction processes.

Fig. 15 demonstrates the 2D-ROC curve of all the comparison methods on the six experimental datasets to compare the superiority of the proposed method from another perspective. As we can see from Fig. 15, except for the Pavia dataset, the proposed method achieved the best results on all experimental datasets. For each fixed PF, the value of the corresponding PD is the highest among all the comparison methods. This superiority is particularly evident on the Hyperion, urban-3, and airport-4 datasets. For a more quantitative comparison, Table I lists the five types of AUC values obtained by all comparison methods to demonstrate the superiority of the proposed method. As we can see from Table I, the values of $AUC_{(PD,PF)}$ of the proposed method are the highest on the experimental datasets among all the comparison methods, and the proposed method has the best $AUC_{(SNPR)}$ value for the HYDIECE, beach-3, urban-3, and the airport-4 datasets. In addition, the $AUC_{(PF,\tau)}$ values are also the best on the HYDICE, beach-3, and the airport-4 datasets, which indicates that the proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

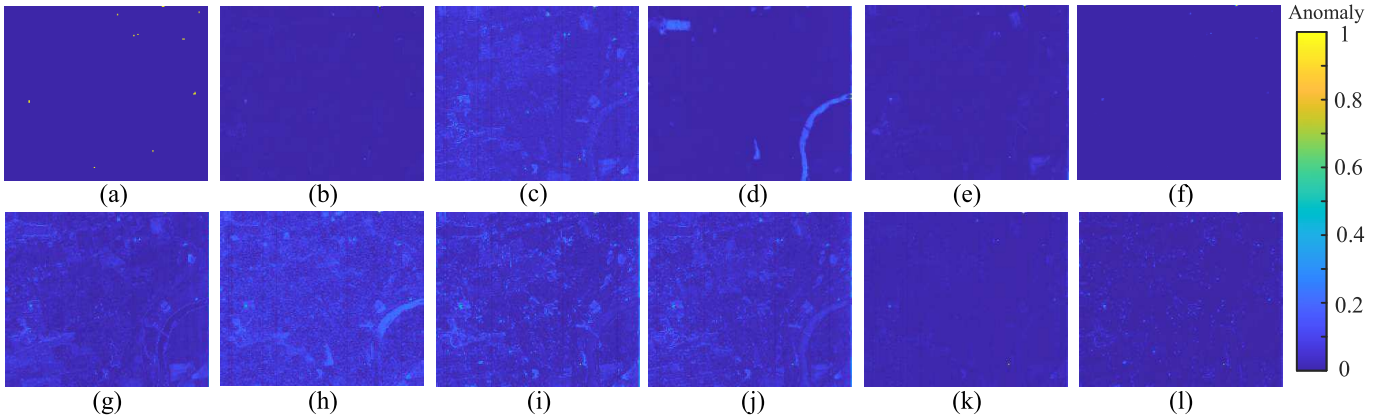


Fig. 9. Detection maps obtained by different methods on the Hyperion dataset. (a) Ground truth, (b) proposed, (c) NLDPAD, (d) LRR-Net-S, (e) DeepLR, (f) MSNet, (g) TAEF, (h) PCA-TLRSR, (i) LARTVAD, (j) GTVLRR, (k) LSMAD, and (l) LRX.

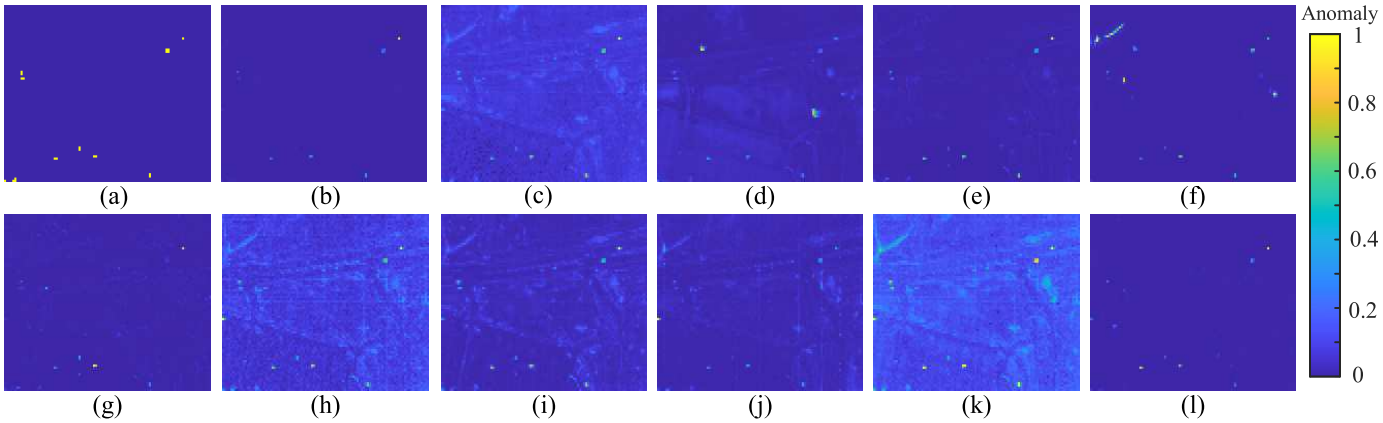


Fig. 10. Detection maps obtained by different methods on the HYDICE dataset. (a) Ground truth, (b) proposed, (c) NLDPAD, (d) LRR-Net-S, (e) DeepLR, (f) MSNet, (g) TAEF, (h) PCA-TLRSR, (i) LARTVAD, (j) GTVLRR, (k) LSMAD, and (l) LRX.

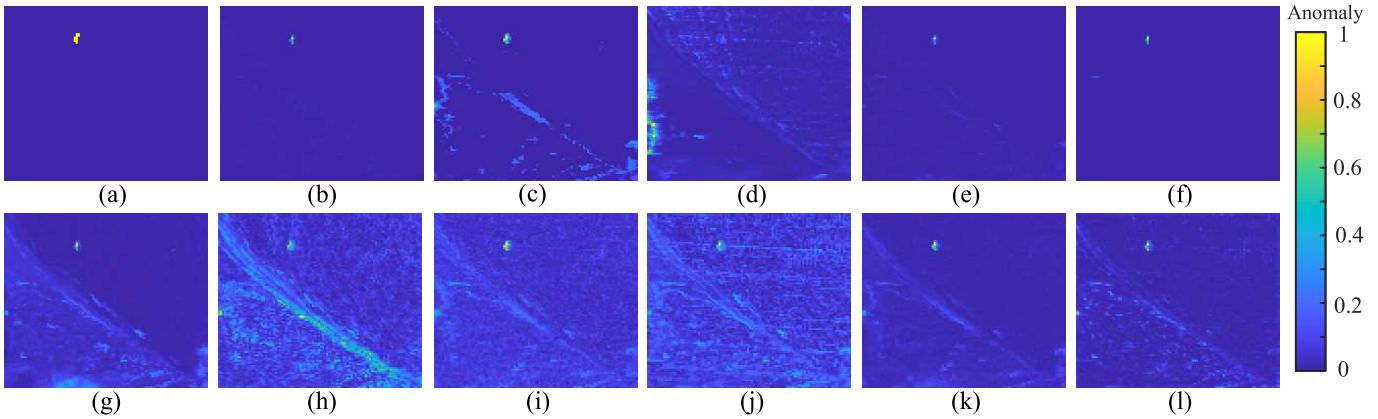


Fig. 11. Detection maps obtained by different methods on the beach-3 dataset. (a) Ground truth, (b) proposed, (c) NLDPAD, (d) LRR-Net-S, (e) DeepLR, (f) MSNet, (g) TAEF, (h) PCA-TLRSR, (i) LARTVAD, (j) GTVLRR, (k) LSMAD, and (l) LRX.

method has the best background suppression performance on those three datasets. The best results of different types of AUC values are marked in black in Table I. Based on the above analysis, we can conclude that the proposed method can suppress the background well while ensuring excellent anomaly detection performance.

Fig. 16 demonstrates the background-anomaly separation maps obtained by all comparison methods on the experimental datasets. In Fig. 16, the red box represents the statistical distribution of anomaly pixel values, and the green box represents the statistical distribution of background pixel values. The larger the interval between those two boxes, the better

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

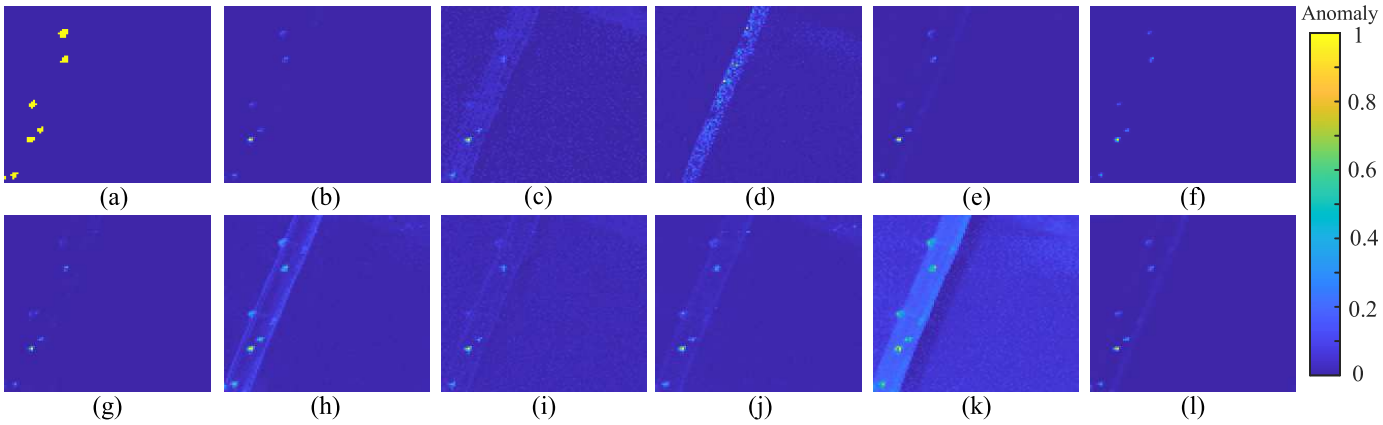


Fig. 12. Detection maps obtained by different methods on the Pavia dataset. (a) Ground truth, (b) proposed, (c) NLDPAD, (d) LRR-Net-S, (e) DeepLR, (f) MSNet, (g) TAEF, (h) PCA-TLRSR, (i) LARTVAD, (j) GTVLRR, (k) LSMAD, and (l) LRX.

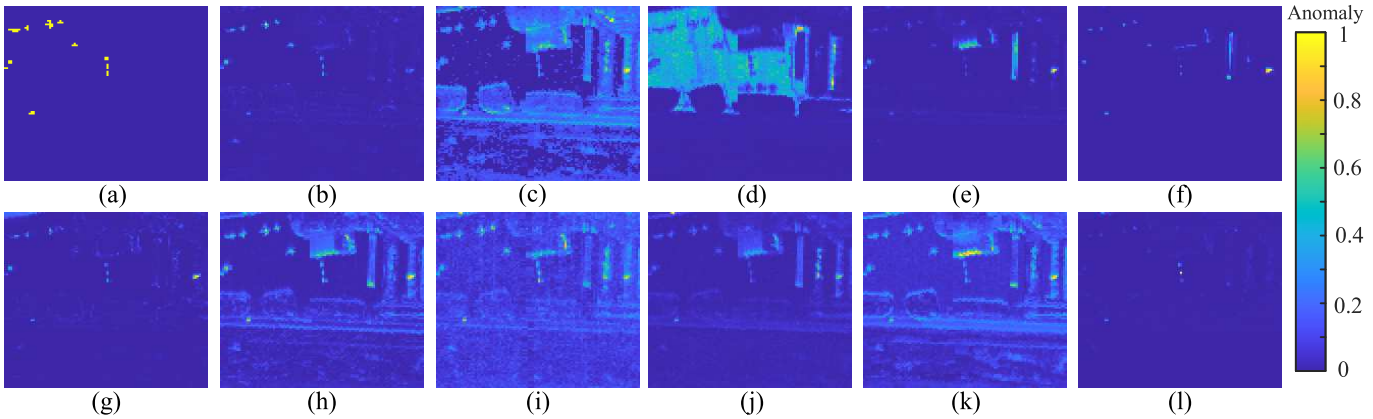


Fig. 13. Detection maps obtained by different methods on the urban-3 dataset. (a) Ground truth, (b) proposed, (c) NLDPAD, (d) LRR-Net-S, (e) DeepLR, (f) MSNet, (g) TAEF, (h) PCA-TLRSR, (i) LARTVAD, (j) GTVLRR, (k) LSMAD, and (l) LRX.

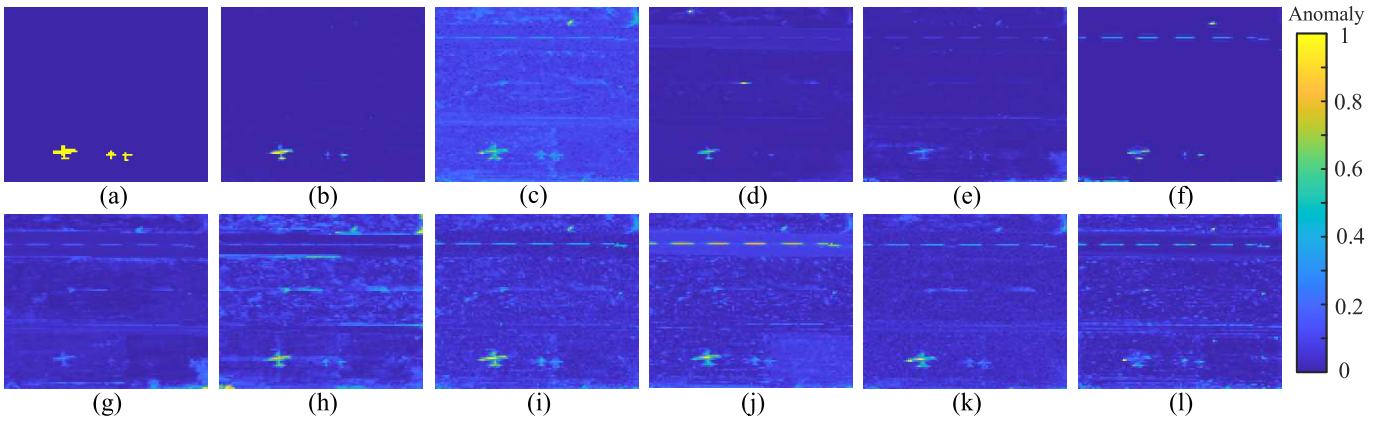


Fig. 14. Detection maps obtained by different methods on the airport-4 dataset. (a) Ground truth, (b) proposed, (c) NLDPAD, (d) LRR-Net-S, (e) DeepLR, (f) MSNet, (g) TAEF, (h) PCA-TLRSR, (i) LARTVAD, (j) GTVLRR, (k) LSMAD, and (l) LRX.

the separability between the anomaly and the background. From Fig. 16, we can see that most model-based methods achieve better separation performance than the deep-learning-based methods among the six experimental datasets. Fig. 16(a) demonstrates the result on the Hyperion dataset, where the LARTVAD wins the best among all comparison methods. The NLDPAD wins the best among the deep-learning-based methods in this dataset. Fig. 16(b) demonstrates the result on the HYDICE dataset, where the LARTVAD wins the best among all comparison methods. The TAEF and the NLDPAD win the best among the deep-learning-based methods. Fig. 16(c) demonstrates the result on the beach-3 dataset, where the LARTVAD and the NLDPAD win the best among all comparison methods, and the proposed method wins the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

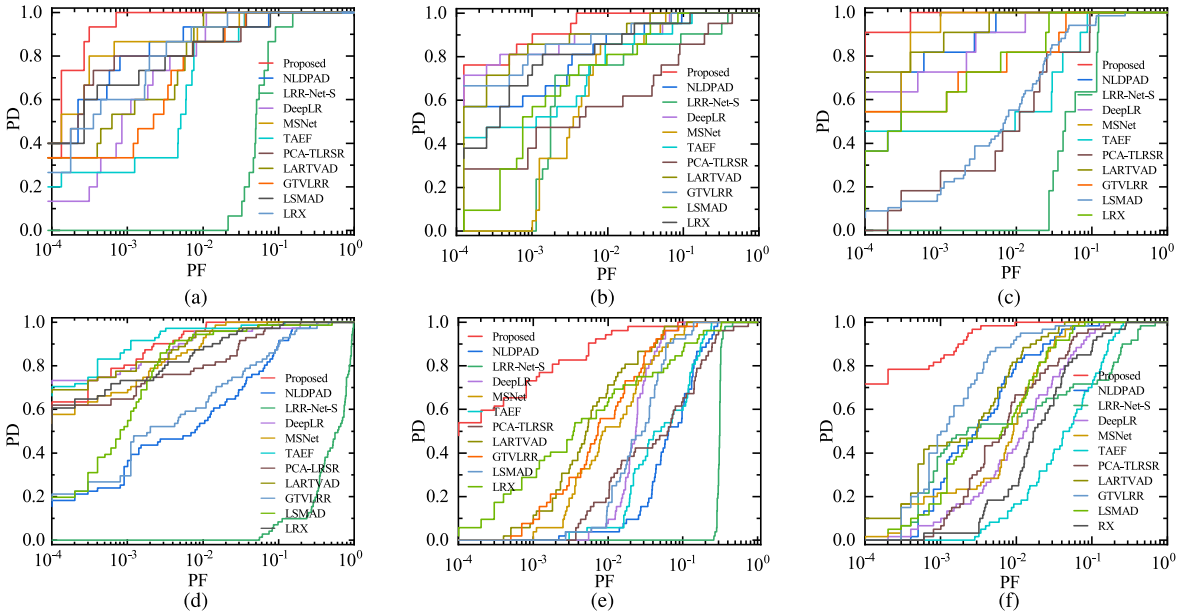


Fig. 15. 2D-ROC curve of all experimental datasets obtained by different comparison methods. (a)–(f) Results of Hyperion, HYDICE, beach-3, Pavia, urban-3, and airport-4 datasets, respectively.

TABLE I

AUC VALUES OF ALL COMPARISON METHODS ON THE EXPERIMENTAL DATASETS

Dataset		Proposed	NLDPAD	LRR-Net-S	DeepLR	MSNet	TAEF	PCA-TLRSR	LARTVAD	GTVLRR	LSMAD	LRX
Hyperion	$AUC_{(PD,PF)}$	**0.9998**	0.9944	0.9409	0.9972	0.9973	0.9923	0.9948	0.9972	0.9944	0.9927	0.9974
	$AUC_{(PD,\tau)}$	0.1994	0.3924	0.0363	0.2147	0.1798	0.3400	0.3921	**0.4035**	0.3550	0.1984	0.2683
	$AUC_{(PF,\tau)}$	0.0046	0.0546	0.0106	0.0074	**0.0006**	0.0374	0.0744	0.0504	0.0446	0.0116	0.0087
	$AUC_{(ODP)}$	1.1946	1.3322	0.9666	1.2045	1.1765	1.2949	1.3125	**1.3504**	1.3047	1.1794	1.2570
	$AUC_{(SNPR)}$	43.0096	7.1821	3.4371	28.9979	**322.9120**	9.0823	5.2715	8.0129	7.9551	17.0342	30.8912
HYDICE	$AUC_{(PD,PF)}$	**0.9995**	0.9927	0.9584	0.9940	0.9904	0.9873	0.9435	0.9971	0.9953	0.9902	0.9917
	$AUC_{(PD,\tau)}$	0.2031	0.4392	0.1852	0.3446	0.2566	0.5148	0.4842	**0.5316**	0.4583	0.2342	0.2612
	$AUC_{(PF,\tau)}$	**0.0019**	0.0735	0.0188	0.0059	0.0033	0.0621	0.0866	0.0910	0.0335	0.0217	0.0026
	$AUC_{(ODP)}$	1.2007	1.3584	1.1249	1.3327	1.2438	**1.4400**	1.3411	1.4377	1.4200	1.2027	1.2504
	$AUC_{(SNPR)}$	**105.2448**	5.9726	9.8645	58.5166	77.6802	8.2866	5.5916	5.8393	13.6617	10.7846	101.9744
beach-3	$AUC_{(PD,PF)}$	**0.9999**	0.9991	0.9326	0.9983	0.9998	0.9756	0.9749	0.9994	0.9916	0.9995	0.9943
	$AUC_{(PD,\tau)}$	0.3759	0.6019	0.0778	0.2740	0.3558	0.4093	0.5008	**0.6590**	0.5349	0.5363	0.4532
	$AUC_{(PF,\tau)}$	**0.0020**	0.0089	0.0272	0.0040	0.0022	0.0350	0.1084	0.0743	0.0999	0.0244	0.0232
	$AUC_{(ODP)}$	1.3739	**1.5922**	0.9833	1.2683	1.3534	1.3499	1.3674	1.5842	1.4266	1.5114	1.4244
	$AUC_{(SNPR)}$	**184.7255**	68.0110	2.8657	68.8203	161.9066	11.6910	4.6209	8.8750	5.3540	21.9775	19.5493
Pavia	$AUC_{(PD,PF)}$	**0.9990**	0.9648	0.4238	0.9912	0.9979	0.9976	0.9856	0.9970	0.9666	0.9902	0.9942
	$AUC_{(PD,\tau)}$	0.1033	0.1469	0.0092	0.1228	0.0904	0.2225	**0.4111**	0.3182	0.1566	0.1919	0.1317
	$AUC_{(PF,\tau)}$	0.0009	0.0131	0.0169	0.0023	**0.0002**	0.0120	0.0652	0.0178	0.0224	0.0141	0.0018
	$AUC_{(ODP)}$	1.1014	1.0986	0.4162	1.1116	1.0881	1.2080	**1.3355**	1.2974	1.1007	1.1680	1.1241
	$AUC_{(SNPR)}$	115.8263	11.2042	0.5465	52.5836	**518.4690**	18.5272	6.3034	17.8969	6.9822	13.5920	72.4428
urban-3	$AUC_{(PD,PF)}$	**0.9968**	0.9063	0.6891	0.9750	0.9774	0.9211	0.8968	0.9867	0.9825	0.9636	0.9628
	$AUC_{(PD,\tau)}$	0.2785	0.3043	0.1088	0.0900	0.0985	0.1100	0.3543	**0.4351**	0.4338	0.1376	0.0784
	$AUC_{(PF,\tau)}$	0.0062	0.1142	0.1190	0.0101	**0.0027**	0.0381	0.1144	0.0599	0.1074	0.0286	0.0035
	$AUC_{(ODP)}$	1.2691	1.0965	0.6790	1.0548	1.0732	0.9930	1.1367	**1.3619**	1.3089	1.0726	1.0376
	$AUC_{(SNPR)}$	**45.1724**	2.6659	0.9145	8.8720	36.6085	2.8869	3.0964	7.2642	4.0401	4.8134	22.1880
airport-4	$AUC_{(PD,PF)}$	**0.9995**	0.9906	0.9129	0.9904	0.9869	0.9284	0.9804	0.9938	0.9956	0.9874	0.9599
	$AUC_{(PD,\tau)}$	0.3851	0.4156	0.1883	0.3495	0.2236	0.1320	0.4566	**0.4906**	0.4191	0.3518	0.2347
	$AUC_{(PF,\tau)}$	**0.0029**	0.1261	0.0126	0.0216	0.0046	0.0401	0.0747	0.0687	0.0561	0.0520	0.0472
	$AUC_{(ODP)}$	1.3816	1.2801	1.0886	1.3183	1.2060	1.0204	1.3622	**1.4157**	1.3586	1.2872	1.1473
	$AUC_{(SNPR)}$	**130.5783**	3.2964	14.9818	16.2089	49.0911	3.2950	6.1086	7.1457	7.4685	6.7659	4.9713

best among the rest of the deep-learning-based methods. Fig. 16(d) demonstrates the result on the Pavia dataset, where the LARTVAD wins the best among all comparison methods and the TAEF wins the best among the deep-learning-based methods, followed by the proposed method. Fig. 16(e) demonstrates the result on the urban-3 dataset, where the proposed method, the LARTVAD, and the GTVLRR win the best among all comparison methods. Fig. 16(f) demonstrates the result on the airport-4 dataset, where the GTVLRR wins the best among all comparison methods. The NLDPAD wins the best among the deep-learning-based methods, followed by the proposed method. From the background-anomaly separation map, it can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

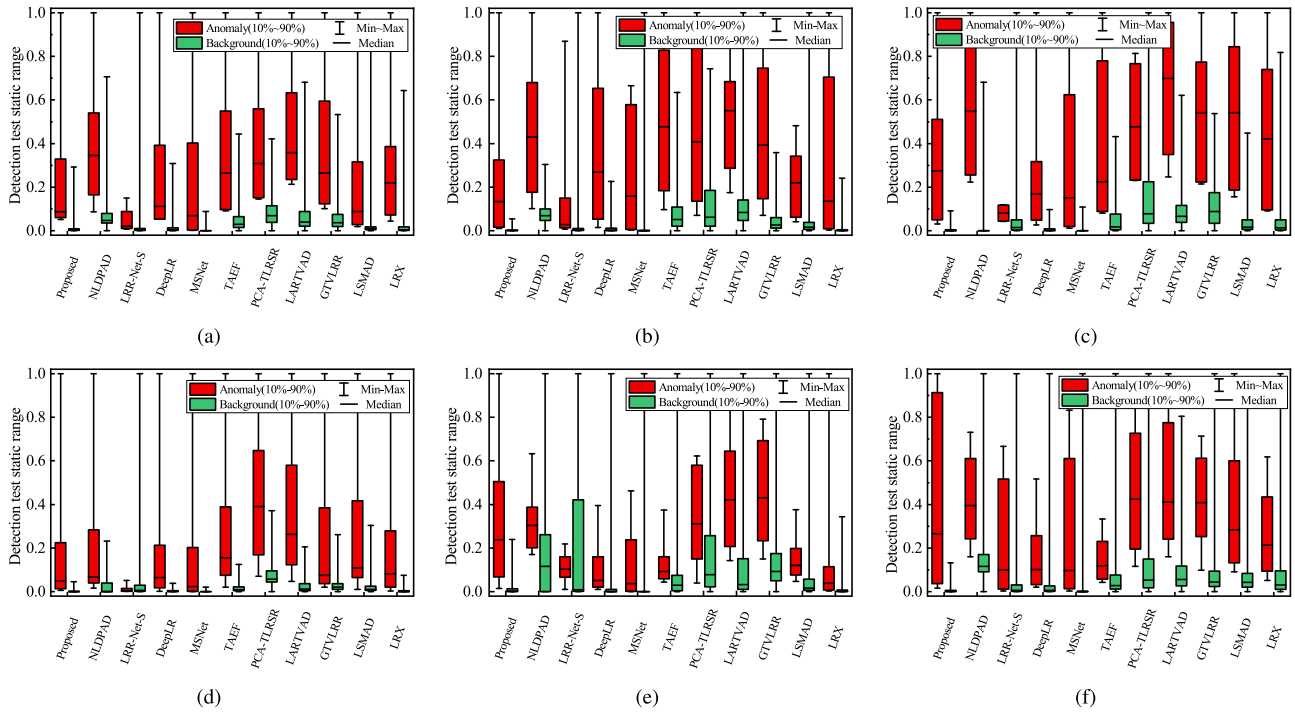


Fig. 16. Background-anomaly separation maps obtained by different comparison methods. (a)–(f) Results of Hyperion, HYDICE, beach-3, Pavia, urban-3, and airport-4 datasets, respectively.

TABLE II
RUNNING TIME (SECOND) OF ALL COMPARISON METHODS ON THE EXPERIMENTAL DATASETS

Dataset	Proposed	NLDPAD	LRR-Net-S	DeepLR	MSNet	TAEF	PCA-TLRSR	LARTVAD	GTVLRR	LSMAD	LRX
Hyperion	13.3460	369.4680	445.4605	10.9784	20.0616	61.7232	12.8480	86.2500	126.1390	**8.8920**	11.8850
HYDICE	34.3782	157.7150	301.1886	4.0624	11.1918	19.9938	8.6240	36.4340	39.7850	**3.4650**	4.8280
beach-3	59.6850	116.7130	355.8480	**4.8699**	10.6524	39.0879	11.6000	19.6150	45.2320	5.3290	6.6290
Pavia	10.5681	142.5250	307.7304	3.2906	10.5555	23.5134	4.5150	21.5450	39.2590	2.7550	**2.7200**
urban-3	83.2578	180.7120	308.0708	**5.0042**	11.6847	25.0508	5.7460	44.1006	41.4420	5.0100	6.7270
airport-4	40.6924	181.0510	316.2721	**4.9018**	10.6936	41.0690	10.8210	33.0010	43.8780	5.6470	6.8700

be seen that the separation performance of deep learning-based methods is generally not as good as that of model-based methods. Nevertheless, the separation performance of the proposed method on the six datasets is relatively advantageous among all the compared deep learning-based methods, except for the NLDPAD.

Table II lists the running time of all comparison methods, which indicates that DeepLR is a relatively computationally efficient detector compared with other methods, while its anomaly detection performance is indeed not very good, which can be seen from the detection maps displayed in Figs. 9–14, the AUC values listed in Table I, and the 2D-ROC curve displayed in Fig. 15. The running time of the proposed method is similar to the MSNet and the TAEF, and much shorter than that of the NLDPAD and the LRR-Net-S, but the proposed method outperforms these four methods in both anomaly detection and background suppression performance, which shows the superiority of the proposed method.

E. Ablation Study

Finally, the following three ablation experiments are carried out to comprehensively validate the effectiveness of the proposed network structure components, the convergence of the training process of the proposed method, and the robustness of the proposed method to noise. First, a componentwise analysis is performed to isolate the contributions of each component in the proposed method, where the proposed method without the MSFE module is marked as "NoMSFE," the proposed method without "LRMF" module is marked as "NoLRMF," the proposed method without "MSFF" module is marked as "NoMSFF," and the proposed method without l_1 regularization is marked as "No"l_1. Table III lists the AUC value results. It can be seen from Table III that when the proposed algorithm lacks any of the components proposed in this article, the performance will be degraded. When all components are included, the performance is outperformed by the detector that lacks one of the components, which fully proves the effectiveness of each component. The second one is carried out to verify the convergence of the proposed method. To simplify the analysis process, we use the loss ς defined in (16) and $\text{AUC}_{(\text{PD,PF})}$ to represent the convergence of the network training process. The loss and $\text{AUC}_{(\text{PD,PF})}$ of the six experiment datasets are displayed in Fig. 18, which indicates that the training loss and the $\text{AUC}_{(\text{PD,PF})}$ of the proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

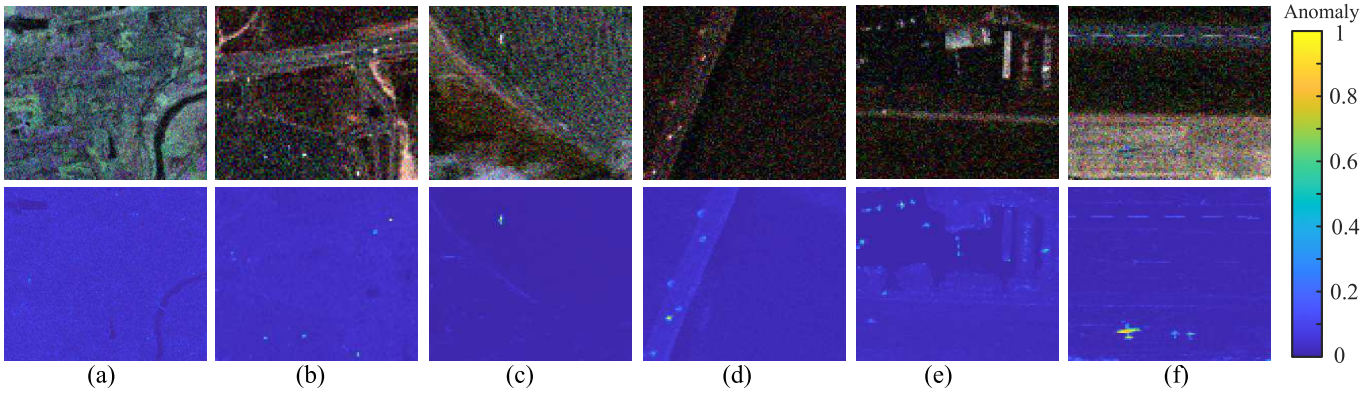


Fig. 17. Anomaly detection maps obtained by the proposed method under the noise case where the noise variance is 0.01. (a)–(f) Results of Hyperion, HYDICE, beach-3, Pavia, urban-3, airport-4 datasets, respectively.

TABLE III

AUC VALUES OBTAINED BY THE PROPOSED METHOD WITH DIFFERENT NETWORK COMPONENTS

Dataset		Pro	NoMSFE	NoLRMF	NoMSFF	Nol_1
Hyperion	$AUC_{(PD,PF)}$	**0.9998**	0.9995	0.9993	0.9995	0.9994
	$AUC_{(PD,\tau)}$	0.1994	0.1730	0.1478	**0.2016**	0.1738
	$AUC_{(PF,\tau)}$	0.0046	0.0045	**0.0041**	0.0048	0.0046
	$AUC_{(ODP)}$	1.1946	1.1680	1.1430	**1.1963**	1.1685
	$AUC_{(SNPR)}$	**43.0096**	38.1839	36.0514	42.2360	37.8186
HYDICE	$AUC_{(PD,PF)}$	**0.9995**	0.9985	0.9987	0.9985	0.9987
	$AUC_{(PD,\tau)}$	0.2031	0.3484	0.2911	**0.3733**	0.3407
	$AUC_{(PF,\tau)}$	0.0037	0.0044	**0.0034**	0.0049	0.0039
	$AUC_{(ODP)}$	1.2007	1.3425	1.2865	**1.3668**	1.3355
	$AUC_{(SNPR)}$	**105.2448**	79.1010	85.2664	75.5333	86.6264
beach-3	$AUC_{(PD,PF)}$	**0.9999**	0.9994	0.9994	0.9993	0.9987
	$AUC_{(PD,\tau)}$	**0.3759**	0.2437	0.2512	0.3725	0.2353
	$AUC_{(PF,\tau)}$	**0.0020**	0.0022	0.0026	0.0031	0.0021
	$AUC_{(ODP)}$	**1.3739**	1.2408	1.2480	1.3687	1.2320
	$AUC_{(SNPR)}$	**123.0016**	111.4880	97.2404	120.4961	114.0036
Pavia	$AUC_{(PD,PF)}$	**0.9990**	0.9986	0.9985	0.9987	0.9985
	$AUC_{(PD,\tau)}$	0.1033	**0.1041**	0.0902	0.0876	0.0986
	$AUC_{(PF,\tau)}$	0.0009	0.0010	0.0009	**0.0008**	0.0009
	$AUC_{(ODP)}$	1.1014	**1.1017**	1.0877	1.0854	1.0961
	$AUC_{(SNPR)}$	**115.8263**	107.1807	98.9466	108.5535	109.2527
urban-3	$AUC_{(PD,PF)}$	**0.9968**	0.9949	0.9935	0.9949	0.9799
	$AUC_{(PD,\tau)}$	0.2785	**0.2795**	0.2656	0.2745	0.1394
	$AUC_{(PF,\tau)}$	**0.0062**	0.0070	0.0069	0.0071	0.0115
	$AUC_{(ODP)}$	**1.2691**	1.2674	1.2523	1.2623	1.1078
	$AUC_{(SNPR)}$	**45.1724**	39.7910	38.6980	38.4012	12.0747
airport-4	$AUC_{(PD,PF)}$	**0.9995**	0.9979	0.9980	0.9980	0.9938
	$AUC_{(PD,\tau)}$	0.3851	**0.4688**	0.4641	0.4436	0.2242
	$AUC_{(PF,\tau)}$	**0.0029**	0.0103	0.0101	0.0084	0.0087
	$AUC_{(ODP)}$	1.3816	**1.4564**	1.4520	1.4332	1.2094
	$AUC_{(SNPR)}$	**130.5783**	45.5278	45.8810	52.7426	25.8722

TABLE IV

$AUC_{(PD,PF)}$ VALUES OBTAINED BY THE PROPOSED METHOD UNDER DIFFERENT NOISE VARIANCES ON THE EXPERIMENTAL DATASETS

	Hyperion	HYDICE	beach-3	Pavia	urban-3	airport-4
$\sigma = 0.001$	0.9998	0.9994	0.9999	0.9964	0.9899	0.9986
$\sigma = 0.005$	0.9997	0.9975	0.9999	0.9982	0.9958	0.9977
$\sigma = 0.01$	0.9997	0.9983	0.9999	0.9948	0..9943	0.9985
$\sigma = 0.05$	0.9975	0.9458	0.9997	0.9939	0.9422	0.9987
$\sigma = 0.1$	0.9781	0.9315	0.9980	0.9888	0.8953	0.9989

method can reach convergence on the six datasets. The third ablation experiment is carried out to verify the robustness of the proposed method to noise. Here, we add Gaussian white noise with a mean of 0 and variance selected from the set $\{0.001, 0.005, 0.01, 0.05, 0.1\}$ to the original HSI data. Fig. 17 shows the noise-poor HSI pseudo-color images and the anomaly detection maps where the noise variance is set to 0.01. It can be seen from Fig. 17 that the proposed method still has a good anomaly detection performance under the noise case. Table IV lists the $AUC_{(PD,PF)}$ values under different noise variances, which further proves the robustness of the proposed method to noise.

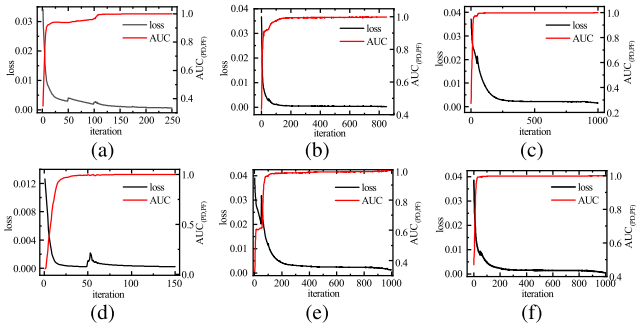


Fig. 18. Loss and $AUC_{(PD,PF)}$ curves on the six experimental datasets. (a)–(f) Results of the Hyperion, the HYDICE, the beach-3, the Pavia, the urban-3, and the airport-4 datasets, respectively.

V. CONCLUSION

In this article, a model-constrained two-stage transform network was proposed for HAD. In the first transformation stage, an MSFE module was introduced to extract the different scale features of the background and transform the HSI from the image domain into the deep domain. Then, a multidirectional tensor LRMF module is proposed to exploit the low rankness of the background in 3-D and guide the first deep transform stage with direction and goal. After that, a Unet with an MSFF module was proposed to fuse the different-level features and transform the first deep feature into another deep domain. Then, the TV regularization was designed on the output of the Unet to exploit the piecewise-smooth property of the background in 3-D and guide the second deep transform stage with direction. All in all, the proposed method employs a divide-and-conquer model constraint strategy, leveraging the low rankness and piecewise smoothness of the background

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

across different deep transform domains to better exploit its potential properties. In turn, the physical properties of the background also guide the deep transforms at different stages, providing a rationale for network structural design and targeted learning. Furthermore, the proposed method employs an MSFE and fusion strategy, enabling better extraction of background features and more accurate background reconstruction. Experiments on six real-world datasets demonstrate the effectiveness of the proposed method.

REFERENCES

[1] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[2] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[3] W. Rao, L. Gao, Y. Qu, X. Sun, B. Zhang, and J. Chanussot, "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.

[4] X. Zhao, K. Liu, K. Gao, and W. Li, "Hyperspectral time-series target detection based on spectral perception and spatial–temporal tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023.

[5] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[6] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Nov. 1990.

[7] H. Kwon, "Adaptive anomaly detection using subspace separation for hyperspectral imagery," *Opt. Eng.*, vol. 42, no. 11, pp. 3342–3351, Nov. 2003.

[8] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[9] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016.

[10] W. Liu et al., "Random selection-based adaptive saliency-weighted RXD anomaly detection for hyperspectral imagery," *Int. J. Remote Sens.*, vol. 39, no. 8, pp. 2139–2158, Apr. 2018.

[11] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641.

[12] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[13] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[14] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[15] Q. Ling, Y. Guo, Z. Lin, and W. An, "A constrained sparse representation model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2358–2371, Apr. 2019.

[16] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[17] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[18] H. Su, Z. Wu, A.-X. Zhu, and Q. Du, "Low rank and collaborative representation for hyperspectral anomaly detection via robust dictionary construction," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 195–211, Nov. 2020.

[19] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[20] T. Guo, L. He, F. Luo, X. Gong, Y. Li, and L. Zhang, "Anomaly detection of hyperspectral image with hierarchical antinoise mutual-incoherence{-} induced low-rank representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023.

[21] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[22] X. Shen, H. Liu, J. Nie, and X. Zhou, "Matrix factorization with framelet and saliency priors for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504413.

[23] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021.

[24] M. Wang, D. Hong, B. Zhang, L. Ren, J. Yao, and J. Chanussot, "Learning double subspace representation for joint hyperspectral anomaly detection and noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023.

[25] J. Wang, Y. Xia, and Y. Zhang, "Anomaly detection of hyperspectral image via tensor completion," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 6, pp. 1099–1103, Jun. 2021.

[26] Z. Chen, B. Yang, and B. Wang, "A preprocessing method for hyperspectral target detection based on tensor principal component analysis," *Remote Sens.*, vol. 10, no. 7, p. 1033, Jun. 2018.

[27] A. Ruhan, X. Mu, and J. He, "Enhance tensor RPCA-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[29] H. Qin, Q. Shen, H. Zeng, Y. Chen, and G. Lu, "Generalized nonconvex low-rank tensor representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023.

[30] S. Sun, J. Liu, and W. Li, "Spatial invariant tensor self-representation model for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 54, no. 5, pp. 3120–3131, May 2024.

[31] X. He, J. Wu, Q. Ling, Z. Li, Z. Lin, and S. Zhou, "Anomaly detection for hyperspectral imagery via tensor low-rank approximation with multiple subspace learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023.

[32] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[33] S. Sun, J. Liu, X. Chen, W. Li, and H. Li, "Hyperspectral anomaly detection with tensor average rank and piecewise smoothness constraints," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8679–8692, Nov. 2022.

[34] S. Sun, J. Liu, Z. Zhang, and W. Li, "Hyperspectral anomaly detection based on adaptive low-rank transformed tensor," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9787–9799, Jul. 2024.

[35] M. Feng, W. Chen, Y. Yang, Q. Shu, H. Li, and Y. Huang, "Hyperspectral anomaly detection based on tensor ring decomposition with factors TV regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.

[36] W. Shang et al., "Hyperspectral anomaly detection via sparsity of core tensor under gradient domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023.

[37] Q. Xiao, L. Zhao, S. Chen, and X. Li, "Hyperspectral anomaly detection via MERA decomposition and enhanced total variation regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–19, 2024.

[38] W. Qin, H. Wang, F. Zhang, J. Wang, X. Cao, and X.-L. Zhao, "Tensor ring decomposition-based generalized and efficient nonconvex approach for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–18, 2024.

[39] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Mar. 2020.

[40] Z. Wu and B. Wang, "Transformer-based autoencoder framework for nonlinear hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508015.

[41] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, and Y. Huang, "Hyperspectral anomaly detection based on spatial–spectral cross-guided mask autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9876–9889, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.

[42] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.
[43] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.
[44] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.
[45] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. J. Plaza, "PDBSNet: Pixel-shuffle downsampling blind-spot reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.
[46] D. Wang, L. Zhuang, L. Gao, X. Sun, X. Zhao, and A. Plaza, "Sliding dual-window-inspired reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, 2024.
[47] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. Plaza, "BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023.
[48] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.
[49] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.
[50] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.
[51] Z. Li, Y. Wang, C. Xiao, Q. Ling, Z. Lin, and W. An, "You only train once: Learning a general anomaly enhancement network with random masks for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023.
[52] K. Li, W. An, Y. Wang, T. Zhang, Y. Qin, and T. Gao, "Explicit background endmember learning for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–17, 2024.
[53] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3631–3645, Feb. 2025.
[54] X. He et al., "Global-to-local spatial–spectral awareness transformer network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–22, 2024.
[55] S.-S. Young, C.-H. Lin, and Z.-C. Leng, "Unsupervised abundance matrix reconstruction transformer-guided fractional attention mechanism for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9150–9164, May 2025.
[56] Z. He, D. He, M. Xiao, A. Lou, and G. Lai, "Convolutional transformer-inspired autoencoder for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[57] Z. Wu et al., "SMCNet: Sparse-inspired masked convolutional network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–17, 2024.
[58] Q. Yan et al., "Efficient image enhancement with a diffusion-based frequency prior," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Mar. 7, 2025, doi: 10.1109/TCSVT.2025.3549351.
[59] X. Ma, K. Shi, D. Chen, Y. Cao, Q. Yan, and Y. Zhang, "A unified framework for deblurring and HDR imaging in dynamic scenes," *IEEE J. Sel. Topics Signal Process.*, early access, Jun. 19, 2025, doi: 10.1109/JSTSP.2025.3581039.
[60] J. Ma, W. Xie, Y. Li, and L. Fang, "BSDM: Background suppression diffusion model for hyperspectral anomaly detection," 2023, *arXiv:2307.09861*.
[61] W. Chen, X. Zhi, S. Jiang, Y. Huang, Q. Han, and W. Zhang, "DWSDiff: Dual-window spectral diffusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–17, 2025.
[62] S. Lin, X. Cheng, Y. Zeng, Y. Huo, M. Zhang, and H. Wang, "Low-rank and sparse representation inspired interpretable network for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.
[63] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.
[64] C. Li et al., "Interpretable networks for hyperspectral anomaly detection: A deep unfolding solution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.
[65] C. Li, B. Zhang, D. Hong, X. Jia, A. Plaza, and J. Chanussot, "Learning disentangled priors for hyperspectral anomaly detection: A coupling model-driven and data-driven paradigm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6883–6896, Apr. 2025.
[66] L. Ren, D. Wang, L. Gao, M. Wang, and M. Huang, "Nonlocal and deep priors for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025.
[67] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Deep low-rank prior for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.
[68] L. Wang, X. Wang, A. Vizziello, and P. Gamba, "RSAAE: Residual self-attention-based autoencoder for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.
[69] S. Lin, M. Zhang, X. Cheng, L. Shi, P. Gamba, and H. Wang, "Dynamic low-rank and sparse priors constrained deep autoencoders for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–18, 2024.
[70] X. Cheng, M. Zhang, S. Lin, Y. Li, and H. Wang, "Deep self-representation learning framework for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.
[71] M. Feng, Y. Zhu, Y. Yang, and Q. Shu, "Deep low-rank and piecewise-smooth constraint tensor model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.
[72] R. Dian, Y. Liu, and S. Li, "Spectral super-resolution via deep low-rank tensor representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 3, pp. 1–11, Mar. 2025.
[73] X. Wang, L. Wang, A. Vizziello, and P. Gamba, "Hyperspectral anomaly detection based on multiscale central difference convolution network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[74] H. Liu, X. Su, X. Shen, and X. Zhou, "MSNet: Self-supervised multiscale network with enhanced separation training for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024.
[75] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, S. Hu, and X. Liu, "Robust multi-stage progressive autoencoder for hyperspectral anomaly detection," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 134, Nov. 2024, Art. no. 104200.
[76] X. Fu, T. Zhang, J. Cheng, and S. Jia, "MMR-HAD: Multiscale mamba reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–14, 2025.
[77] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.
[78] Z. Wen, W. Yin, and Y. Zhang, "Solving a low-rank factorization model for matrix completion by a nonlinear successive over-relaxation algorithm," *Math. Program. Comput.*, vol. 4, no. 4, pp. 333–361, Dec. 2012.
[79] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.
[80] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:09:54 UTC from IEEE Xplore. Restrictions apply.